Mathematics Bootcamp

Functions, Counting, The Real Numbers

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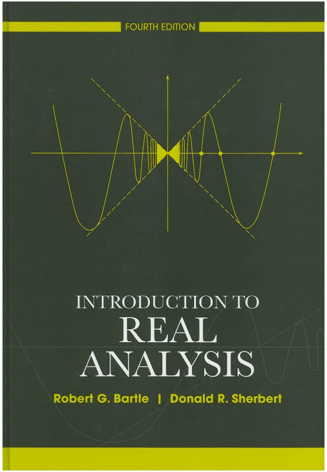
Today

- Cartesian Products and Functions
- Finite and Infinite Sets
- The Real Numbers: Order and Completeness
- Distance, Sequences, and Compactness
- Ordinal Utility

Bartle-Sherbert - Real Analysis



- Most of what follows is taken from Bartle-Sherbert.
- My favorite introductory book for Graduate Level Math.
    - Careful attention to detail.
    - Great pedagogical order.

Cartesian Product

1.1.5 Definition

For nonempty sets A and B, the **Cartesian product** is

$$A \times B := \{(a, b) : a \in A,\ b \in B\}.$$

If $A := \{x : 1 \leq x \leq 2\}$ and $B := \{y : 0 \leq y \leq 1 \text{ or } 2 \leq y \leq 3\}$, the product is not a rectangle.

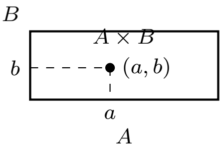


Figure 1.1.2

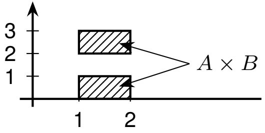


Figure 1.1.3

Functions

1.1.6 Definition

Let A and B be sets. Then a **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$.

$D(f) := A$ is the **domain**; $R(f)$, the set of second elements, is the **range**, with $R(f) \subseteq B$.

Functions as Graphs

- A function f maps a to *exactly one* b: $f(a) = b$ or $a \mapsto b$

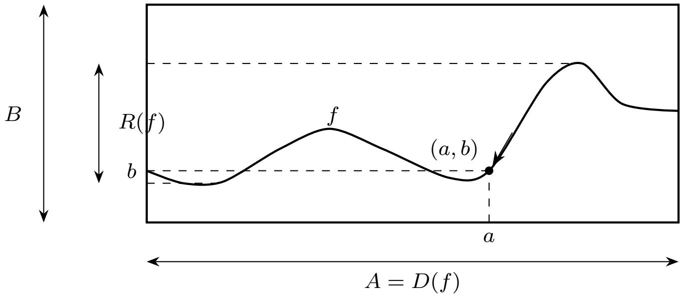


Figure 1.1.4 A function as a graph

Injective, Surjective, and Bijective

1.1.9 Definition

Let $f : A \to B$ be a function.

(a) f is **injective** (**one-to-one**) if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.

(b) f is **surjective** (maps A **onto** B) if $f(A) = B$; that is, if $R(f) = B$.

(c) f is **bijective** if it is both.

Such an f is called an *injection*, a *surjection*, or a *bijection*.

Proofs - Injective and Surjective

- In order to prove that a function f is injective, we must establish that:

$$\text{for all } x_1, x_2 \text{ in } A, \quad \text{if } f(x_1) = f(x_2), \text{ then } x_1 = x_2.$$

  To do this we assume that $f(x_1) = f(x_2)$ and show that $x_1 = x_2$.

- To prove that a function f is surjective, we must show that for any $b \in B$ there exists at least one $x \in A$ such that $f(x) = b$.

Composition and Inverse Images

1.1.12 Definition

If $f : A \to B$ and $g : B \to C$, the **composite function** $g \circ f : A \to C$ is defined by

$$(g \circ f)(x) := g\big(f(x)\big) \qquad \text{for all } x \in A.$$

1.1.7 Definition

For $H \subseteq B$, the **inverse image** of H under f is

$$f^{-1}(H) := \{x \in A : f(x) \in H\}.$$

Cardinality - Finite and Infinite Sets

1.3.1 Definition

(a) $\varnothing$ has 0 **elements.**
(b) S has n **elements** if there is a bijection $\mathbb{N}_n := \{1, 2, \ldots, n\} \to S$.
(c) S is **finite** if it is empty or has n elements for some $n \in \mathbb{N}$; **infinite** otherwise.

1.3.6 Definition

(a) S is **denumerable** (**countably infinite**) if there is a bijection $\mathbb{N} \to S$.
(b) S is **countable** if it is finite or denumerable; **uncountable** otherwise.

Cardinalities of Infinite Sets

Many of our intuitions, and even formal notions, break at infinity.

Comparing Sizes

Two sets have the same cardinality when there is a bijection between them. This definition applies to both finite and infinite sets.

- For finite sets, cardinality counts the elements; for infinite sets, it grades the size of the infinity.
- $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countably infinite: each can be put in bijection with $\mathbb{N}$.
- $\mathbb{R}$ is uncountable: there is no bijection $\mathbb{N} \to \mathbb{R}$.

Cantor's diagonal argument proves the second claim: video.

The Role of the Real Numbers

- Economic alternatives often lie in $\mathbb{R}^L$, while utility is real-valued.
- We need the structure of $\mathbb{R}$, not a construction of it.
- Three ingredients will matter:
    - order, for comparisons;
    - completeness, for limits and extrema;
    - distance, for balls and convergence.

$\mathbb{R}$ Is a Complete Ordered Field

Ordered Field

A **field** supports addition and multiplication. A **total order** compares every pair and is compatible with those operations.

$$a < b \implies a + c < b + c,$$
$$0 < a \text{ and } 0 < b \implies 0 < ab.$$

Totality gives **trichotomy**: exactly one of $a < b$, $a = b$, $a > b$ holds.

Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. Only $\mathbb{R}$ is **complete**: it has no gaps.

Absolute Value and the Real Line

2.2.1 Definition

For $a \in \mathbb{R}$, its **absolute value**, denoted $|a|$, is

$$|a| := \begin{cases} a, & \text{if } a > 0, \\ 0, & \text{if } a = 0, \\ -a, & \text{if } a < 0. \end{cases}$$

Distance on the real line: $|a - b|$.

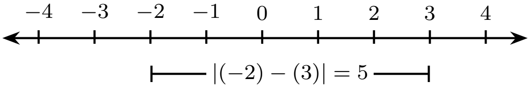


Figure 2.2.3 The distance between $a = -2$ and $b = 3$

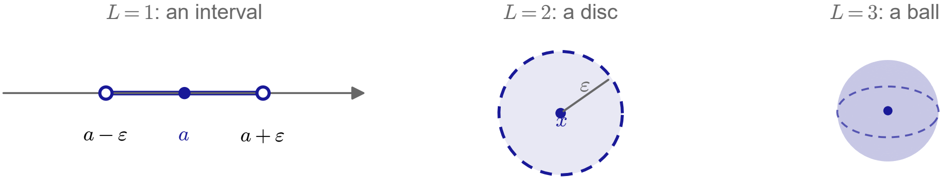
How does a neighborhood look in two dimensions, in three, in more than three?

The Open Ball

Open Ball

For $x \in \mathbb{R}^L$ and $\varepsilon > 0$,

$$B_\varepsilon(x) := \{y \in \mathbb{R}^L : \|y - x\| < \varepsilon\}, \qquad \|z\| := \sqrt{\textstyle\sum_{\ell=1}^{L} z_\ell^2}.$$



On the real line $\|z\| = |z|$, so $B_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$: the leftmost panel, which Bartle calls the ε-neighborhood $V_\varepsilon(a)$ (2.2.7 Definition). Same set, one name in every dimension: the open ball.

Bounds

2.3.1 Bounds

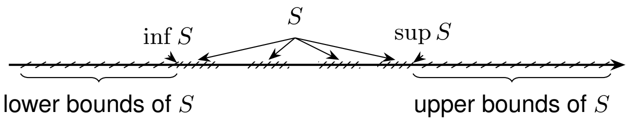
Let $\varnothing \neq S \subseteq \mathbb{R}$.

$$\begin{aligned} u \text{ is an upper bound} \quad &\iff \quad s \leq u \quad \text{for every } s \in S, \\ w \text{ is a lower bound} \quad &\iff \quad w \leq s \quad \text{for every } s \in S. \end{aligned}$$

S is **bounded** if it has both an upper and a lower bound.

Example. For $S = (0, 1)$: upper bounds $= [1, \infty)$, lower bounds $= (-\infty, 0]$.

Suprema and Infima



2.3.2 Supremum = least upper bound

$$u = \sup S$$

1. $s \leq u$ for every $s \in S$.
2. $u \leq v$ for every upper bound v.

$$u \in S \Rightarrow u = \max S;$$
$u \notin S \Rightarrow$ no maximum.

Infimum = greatest lower bound

$$w = \inf S$$

1. $w \leq s$ for every $s \in S$.
2. $t \leq w$ for every lower bound t.

$$w \in S \Rightarrow w = \min S;$$
$w \notin S \Rightarrow$ no minimum.

Completeness of the Real Numbers

2.3.6 Completeness Property of $\mathbb{R}$

For every nonempty $S \subseteq \mathbb{R}$,

$$S \text{ bounded above} \quad \Longrightarrow \quad \sup S \in \mathbb{R}.$$

Consequently, if S is bounded below, then $\inf S \in \mathbb{R}$.

Why $\mathbb{Q}$ is not complete. $A = \{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above, but its supremum in $\mathbb{R}$ is $\sqrt{2} \notin \mathbb{Q}$.

Key Implications for Proofs

Archimedean Property

2.4.3 *If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.*

2.4.5 Corollary *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \varepsilon$.*

Density Results

2.4.8 Density Theorem *If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.*

2.4.9 Corollary *If x and y are real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.*

The Triangle Inequality Controls Distances

Triangle Inequality

For every $x, y, z \in \mathbb{R}^L$,

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

Sequence Convergence in $\mathbb{R}^L$

Convergence

A sequence (x_n) in $\mathbb{R}^L$ **converges** to $x \in \mathbb{R}^L$ if

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N, \qquad \|x_n - x\| < \varepsilon.$$

We write $x_n \to x$.

Coordinatewise Convergence

For $x_n = (x_{n1}, \ldots, x_{nL})$ and $x = (x_1, \ldots, x_L)$,

$$x_n \to x \quad \iff \quad x_{n\ell} \to x_\ell \quad \text{for every } \ell = 1, \ldots, L.$$

Continuity of a Function

Sequential Definition

Let $X \subseteq \mathbb{R}^L$, $Y \subseteq \mathbb{R}^K$, and $f : X \to Y$. The function f is *continuous at* $x \in X$ if, for every sequence (x_n) in X,

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x).$$

It is continuous on X if it is continuous at every $x \in X$.

Keep the objects separate. This is continuity of a *function*. Day 3 defines continuity of a *preference relation*.

Open, Closed, Bounded

Sets in $\mathbb{R}^L$

Let $D \subseteq \mathbb{R}^L$.

- D is **open** if every $x \in D$ has some $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq D$.
- D is **closed** if its complement $\mathbb{R}^L \setminus D$ is open.
- D is **bounded** if $D \subseteq B_M(0)$ for some $M > 0$.

All three rest on the same measurement: the ball $B_\varepsilon(x)$, and so the distance $\|y - x\|$.

Compactness in $\mathbb{R}^L$

Heine–Borel and Extreme Values

Let $D \subseteq \mathbb{R}^L$. Then

$$D \text{ is compact} \quad \Longleftrightarrow \quad D \text{ is closed and bounded.}$$

If D is nonempty and compact and $f : D \to \mathbb{R}$ is continuous, then there exist $x_{\min}, x_{\max} \in D$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \qquad \text{for every } x \in D.$$

Ordinal Utility

Ordinal Utility

A function $u : X \to \mathbb{R}$ represents $\succeq$ if

$$x \succeq y \quad \Longleftrightarrow \quad u(x) \geq u(y).$$

If $\phi : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences.

Existence. On finite or countable X, rationality suffices. On $\mathbb{R}^L_+$ it does not: continuity is needed too.

Back to Economics - Utility Functions

Utility Functions

In economics, we often describe preference relations by means of a *utility function.* A utility function $u(x)$ assigns a numerical value to each element in X, ranking the elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \to \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Utility Functions are Real-Valued Functions

Exercise A (Archimedean Property)

Let $u(x, y) = x^{\alpha} y^{1-\alpha}$ with $0 < \alpha < 1$ and $(x, y) \in \mathbb{R}^2_{++}$. Show that there exists $n \in \mathbb{N}$ such that

$$u(nx, ny) > 1.$$

Exercise B (Density of $\mathbb{Q}$)

Let $u(x, y) = x + y$ and consider the level set $u = 1$. Show that there are infinitely many rational bundles $(r, s) \in \mathbb{Q}^2_{+}$ with $u(r, s) = 1$.

The two completeness implications, at work on economic objects.

Utility Representation Implies Rationality

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Representation and Choice

Representation Preserves Maximizers

Suppose $u : X \to \mathbb{R}$ represents $\succeq$. For every nonempty menu $B \subseteq X$ on which a best alternative exists,

$$C_{\succeq}(B) = \{x \in B : x \succeq y \text{ for every } y \in B\} = \arg\max_{x \in B} u(x).$$

If $\phi : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then

$$\arg\max_{x \in B} u(x) = \arg\max_{x \in B} (\phi \circ u)(x).$$